

Anyuta Kaminskaya 09/21/2026

This program sells a limited number of cinema tickets at a time and keeps track of the number of buyers. The user can only buy 4 tickets at a time until 0 tickets remain. Then the number of buyers is stated.

Functions in this program:

- `buy_tick(ticket_available, buyers):`

Brief Description: Repeatedly asks user to enter how many tickets they want to buy 1-4, until 0 tickets remain. It calls the `count(buyers)` function to keep track of all valid inputs which is then displayed as the total buyers.

Parameters: Parameters include “tik_available” and “buyers”. There are a limited number of tickets available which is 20 and the initial number of buyers is set to 0.

Data Type in Variables:

- `user_input` → integer
- `tick_available` → integer
- `buyers` → integer

Logical Steps of Func:

1. A while loop keeps the function running as long as the value of tickets in `tik_available` haven't run out.
2. The user is asked how many tickets they want to buy and that integer is named `user_input`.
3. To account for any user error, such as entering anything but a positive integer between 1-4, or buying more tickets than available, if else statements are put in place.
4. Then the number of tickets the user buys is subtracted from the available tickets. This new value is the remaining amount of tickets and becomes the new value for `tick_available`.
5. The `count(buyer)` function is called which adds a 1 to buyers.
6. The loop continues until all tickets are sold.
7. Then the user is notified and the number of buyers is stated.

- count(buyers):

Brief Description: Counts the amount of buyers when called upon in buy_tick() function.

Parameters: Parameters include “buyers” which is set to zero.

Data Type in Variables:

- buyers → integer

Logical Steps of Func:

1. The function can only be called if a valid integer was entered within the buy_tick() function.
2. So1 is added to the assigned count for buyers.
3. Then the value for buyer is returned.
4. This repeats until all the tickets are sold.

Return: This function returns the accumulated integer for buyer.

PythonCOP2373 master

Project
PythonCOP2373
venv
AnyutaKaminskaya_ProgramminExcercise#1.py
COP2373.py
TestFile.py
External Libraries
Scratches and Consoles

AnyutaKaminskaya_ProgramminExcercise#1.py

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def buy_tick(tick_available, buyers):
    # count() is called to add 1 to buyers.
    else:
        tick_available -= user_input
        buyers = count(buyers)
        print(f'{tick_available} cinema tickets remaining!')

    # When loop ends, the user is notified that they cannot buy anymore and
    # the number of buyers is stated.
    print(f'Sorry we sold out! Total Buyers: {buyers}.')

# Defining count() function with parameters so buyers start off as 0
def count(buyers):
    # One is added to buyers each time function is called
    buyers += 1
    # The integer assigned to buyers is returned so it can be used when called.
    return buyers

# Calling function and assigning data to tick_available and buyer.
buy_tick(tick_available=20, buyers=0)
```

Run
AnyutaKaminskaya_ProgramminExcercise#1

```
44 Defining tickets remaining:
How many tickets would you like to buy?: 4
8 cinema tickets remaining!
How many tickets would you like to buy?: 4
4 cinema tickets remaining!
How many tickets would you like to buy?: 3
1 cinema tickets remaining!
How many tickets would you like to buy?: 1
0 cinema tickets remaining!
Sorry we sold out! Total Buyers: 6.

Process finished with exit code 0
```

PythonCOP2373 > AnyutaKaminskaya_ProgramminExcercise#1.py

53/67 LF UTF-8 4 spaces Types: Built-in venv [3.14.7]